

Students

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Introduction

The Advanced Goalkeeper system is an AI powered solution for real time ball tracking using computer vision. It overcomes the limitations of manual observation by proposing an automated system that uses a small-scale miniature goal model for evaluation.

Key Objectives

- **Lightweight AI:** Create an optimized deep learning model (MobileNetV2 TFLite) for efficient inference on Raspberry Pi 5.
- **Environmental Robustness:** Eliminate background bias by utilizing a large-scale randomized dataset.
- **Predictive Defense:** Track trajectories and predict future ball landing zones using a mathematical Kalman Filter.

Methodology

Hardware Setup

- Processing: Raspberry Pi 5.
- Vision: Official Camera Module (30-60 FPS).
- Actuation: Dual High-Torque DS3225MG Servos for rapid paddle positioning.

Software & Detection Model

- **Dataset:** Massive collection of 20,000+ images captured across diverse lighting and cluttered environments.
- **Architecture:** 5-zone classification model (Left, Center, Right, etc.) built with MobileNetV2 for maximum classification speed.

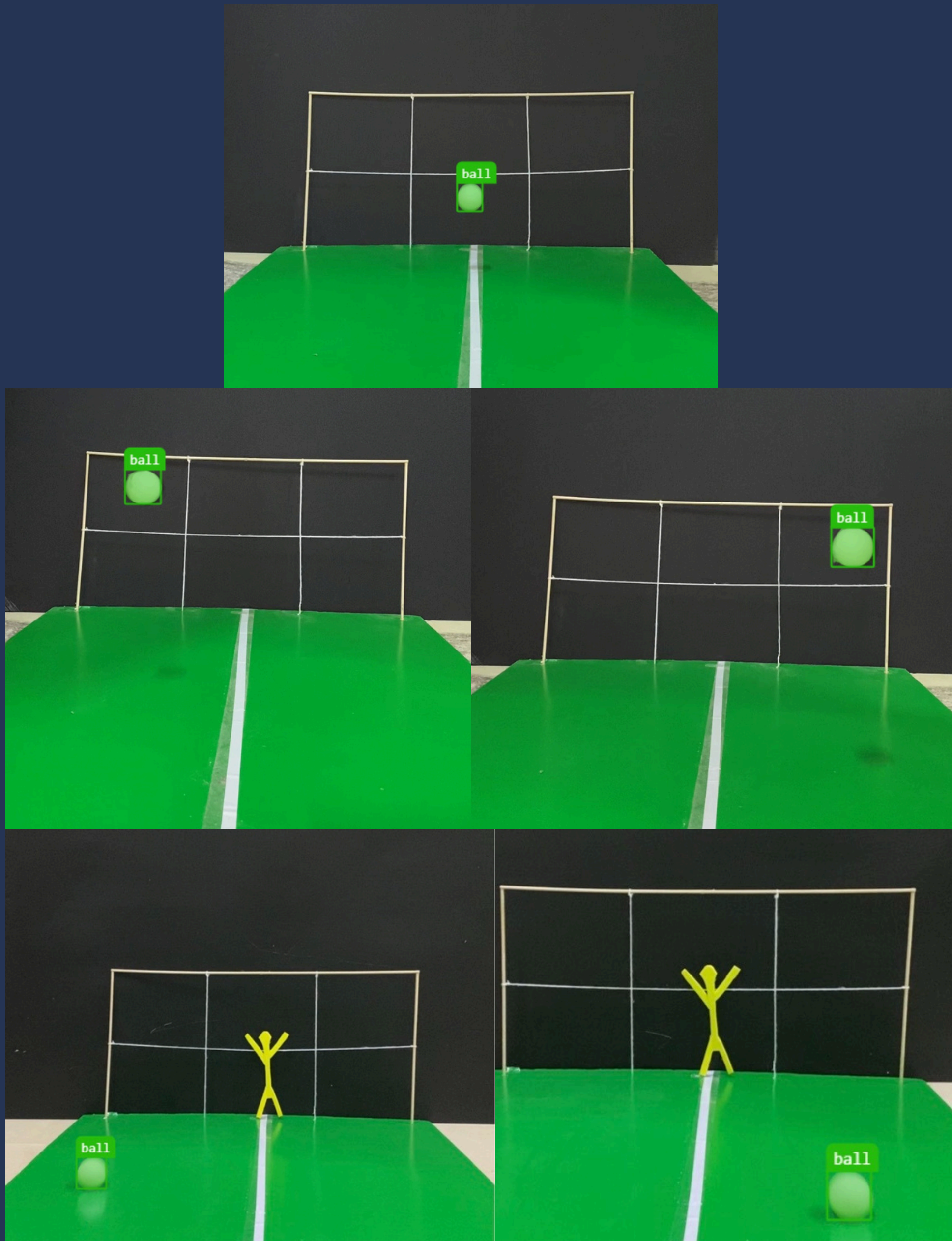
Experiments

Performance Targets:

- Processing Speed: 30+ FPS (real-time).
- Latency: <30 ms per frame.
- Minimum Classification Accuracy: 90%.

Testing Parameters:

- Live testing on hardware (Raspberry Pi 5) under varying lighting and distances.
- Metrics: Inference accuracy, prediction error, and total reaction time.



Conclusion

Live hardware tests demonstrated stable, real-time detection and actuation even under challenging lighting, proving the viability of deploying optimized classification models on edge devices. The miniature field concept provides a solid framework for future sports analytics and high-speed robotic applications.